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Why Earth is Special :

- It is present in habitable zone (range of distances from a star over which water can remain in a liquid state). Habitable zone is also called Goldilocks zone. If it was closer to Sun, then it would have been too hot or vice versa.

- Earth is the right size for sustaining life, this size allows it to have the gravity which holds our atmosphere & allows survival.

- Magnetic field is also an important factor as it protects us from high- ~~speed~~ energy particles coming & hitting Earth from Space called cosmic rays. It also protect us from solar winds. These particles can harm life by letting UV ray in atmosphere & destroying ozone layer.

- Oxygen is also important because it allows respiration in organisms & combustion on Earth. But, there is a 3 atom oxygen molecule which serves to protect us from harmful UV rays. That particle is known as Ozone (O_3)

Why venus is the most hot ?

- Mercury is the closest to Sun but Venus is the most hot, this is because atmosphere of Venus consists mostly of CO_2 which does not let heat escape & keeps the planet hot. This effect is known as Greenhouse effect

- Greenhouse effect is present on Earth too. Though CO_2 in Earth's atmosphere constitute to 0.04% only, the heat ~~trap~~ trapped by the ~~effect~~ effect is enough to keep Earth habitable & water in liquid state.

Why Mars ?

- For scientists, Mars has been a important & interesting topic as it lies just at the edge of Sun's habitable zone, but has no traces of life. But, scientists think that ~~a~~ in the past, there may have been source of liquid water &

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conditions supporting life.

- ISRO also launched a satellite for exploring Mars called Mangalyaan in 2013. It was a smart & low-cost tech. satellite made for ~~reo~~ research on Mars.

Spheres

Hydrosphere

Refers to the water covering 70% of Earth present in diff. water bodies. It is important for plants, marine life & humans. It is also important for evolution. It influence life on land too.

Geosphere

The ~~sol~~ solid part of Earth consisting of soil, rocks & minerals is called geosphere. Crust, mantle & core, all are a part of this. Soil is rich in nutrients (nitrogen & ~~K~~ potassium) & is suitable for ~~fam~~ farming.

Biosphere

The zone of Earth where air, land & water interact with each other to make life possible. All living beings live in this zone & ecosystems 're a unit of this.

Diversity

Geodiversity

The variety of landforms, Earth's rocks, minerals, soil & geological processes is known as Geodiversity.

Biodiversity

Variety of living organisms, species, genes & ecosystem found on Earth.

Ecological balance is the stable interaction b/w living organisms & environment. It's important because everything depends on environment & disruption in this balance can cause big problems like loss of biodiversity ~~loss~~ & climate change.

Reproduction

- Reproduction is the process that ensure continuity of existence of each type of organism & life on Earth.

↳ Genes : Genetic materials or genes is ~~are~~ the substances with stores & passes biological info. from generation to another. More specifally, gene is a segment of DNA (Genetic material).

Asexual reproduction	Sexual reproduction
1) Requires only 1 parent	Requires 2 parents.
2) The offspring is identical to the parent.	The offspring might have feature similar to parents, but is not identical to them
3) Asexual reproduction can be divided into: 1) Vegetative propogation (Eg - Potato) 2) Budding (Eg - Yeast) 3) Fragmentation (Eg - Spirogrya) 4) Spores (Eg - Fern) 5) Binary Fission (Eg - Bacteria)	In this two parents're involved called male & female. Both release ~~Spl~~ spl. cells called gametes (male = sperm; female = egg) which fuse together zygote (fertilized cell capable of starting a new life) Gametes carry chromosomes (genetic info.).

In Depth : Sexual Reproduction

- Both gametes have 23 chromosomes each & get mixed to give the off-spring a unique set of features.

- We know pollination, lets study what happens after that. The pollen grain releases male gametes that fuse with female gamete to form zygote, this process is called fertilization & as the zygote develops, the ovary becomes the fruit. Agent that helped in pollination, can help in seed dispersal also.

- Aquatic animals (except dolphin) reproduce by first releasing sperm & egg in water, & when both gametes combine, they form zygote which turns ~~of~~ into embryo in water too.

- In birds & mammals, zygote formation has the same process, sperm is deposited inside the body of female & the sperm swims towards the egg, & forms zygote.

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- In most mammals, embryo growth happens inside mother's body. Embryo takes all O_2 & nutrient through ~~mo~~ mother's body. This duration is usually called pregnancy. But, in bird, mother usually lays an egg which has zygote & all the food needed for growing baby bird is already present in the egg.

Threats to Life

- There are 3 major threat that the whole world is currently dealing from : climate change, biodiversity loss & pollution. They all are interlinked

- Burning fossil fuel & pollution from different industries cause a rise in pollution & increases carbon & methane ~~CFC gases~~ in atmosphere. These gases lead to the increase of heat being traping causing global warming. Naturally Earth balance this out & absorbs CO_2 but these gases in excess & deforestation reduce this ability of Earth.

- Long-term changes in temp., rainfall & weather patterns is known as climate change. It also affects water supply, human health & wildlife habitat

- Biodiversity is imp. because diverse ecosystems are stronger & more balanced.

- Everyone can help by simple preventive measures Eg- 3Rs

Fact

Though the core is very hot, metals there, exists in a solid state.